

Gyuwon Na

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EDUCATION

Keimyung University | Daegu, South Korea

KMU-DigiPen Special dual-degree program | GPA: 4.36 / 4.5

Mar 2023 – Jun 2026

Relevant Coursework: Computer Graphics, Data Structures, Parallel Programming, Game Engineering, Computer Vision

TECHNICAL SKILLS

Programming Languages: C++, C, Python, JavaScript, GLSL, C#

Frameworks & Libraries: OpenCV, PyTorch, CUDA, OpenGL, WebGL, Unity, ROS 1/2

Tools & Systems: Linux (Ubuntu), CMake, Git/GitHub

Interpersonal Skills: Cross-functional Collaboration, Technical Communication, Problem Solving, Adaptability

RESEARCH EXPERIENCE

ISIP Lab | Undergraduate Student Researcher | Daegu, South Korea

Aug 2025 - Present

- Implemented an adaptive binarization-based lane detection pipeline using OpenCV for robust lane segmentation under varying lighting conditions.
- Developed a rotation-robust image matching framework for challenging image pairs with large viewpoint and orientation changes.
- Conducted literature reviews, failure-case analysis, and reproducible experiments while developing research communication and documentation skills.

PROJECTS

STAND-BY | Technical Leader

Mar 2024 – Jun 2024

- Collaborated with a team of 4 members to implement core gameplay systems, including order processing, scoring logic, customer interactions, and game flow.
- Managed code integration, build testing, debugging, and milestone delivery throughout the development cycle.
- Received the Best Project Award at the Keimyung University Micro-Degree Expo.

MANZO | Technical Leader

Jul 2024 – Jun 2025

- Led technical development of a rhythm-based horror adventure game built with a custom C++ / OpenGL game engine.
- Implemented a custom vector-font rendering system using OpenGL vertex and fragment shaders for in-game text and UI rendering.
- Implemented AI behaviors such as flocking, obstacle avoidance, and autonomous agent movement while maintaining build stability across milestones.

CREATIVITY MOBILITY COMPETITION | Computer Vision Team Leader

Aug 2025 – Oct 2025

- Represented the computer vision team in a mobility competition focused on autonomous perception and intelligent vehicle systems.
- Developed computer vision modules for visual perception tasks such as object detection, scene understanding, and navigation support.
- Coordinated integration between the computer vision, hardware, and control teams to support system-level testing.
- Conducted debugging and performance testing under competition constraints to improve reliability of the perception pipeline.

KAIST MOBILITY CHALLENGE COMPETITION | COMPUTER VISION / PERCEPTION ENGINEER

Dec 2025 – Feb 2026

- Developed perception components for an autonomous mobility challenge involving camera-based environment understanding and navigation.
- Implemented computer vision pipelines using OpenCV and ROS for processing sensor data and supporting real-time decision-making.
- Integrated perception outputs with planning and control modules to improve autonomous driving performance during testing.
- Debugged sensor calibration, perception errors, and system integration issues in preparation for challenge evaluation.

LEADERSHIP EXPERIENCE

Keimyung University | Class Representative | DAEGU, SOUTH KOREA

- Served as the primary communication channel between faculty members and students.
- Organized academic events, student activities, and departmental programs for over XX students.
- Coordinated participation in industry events including G-STAR and student networking activities.
- Facilitated communication, conflict resolution, and collaboration among students and faculty.